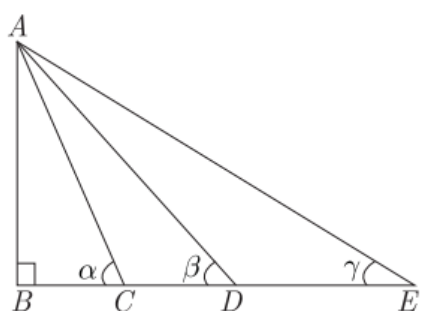


Chapter

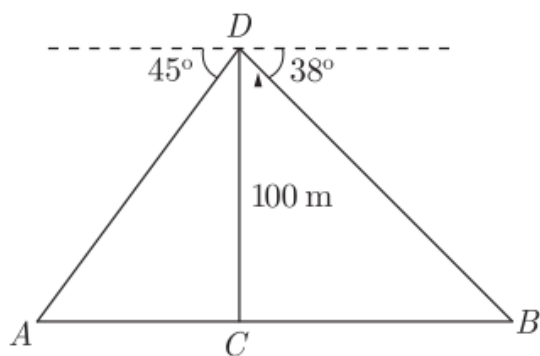
Height And Distance

1. An observer at point E, which is at a certain distance from the lamp post AB, finds the angle of elevation of top of lamp post from positions C, D, and E as α , β , and γ . It is given that B, C, D and E are along a straight line. Which of the following condition is satisfied? SQP 2024



- (a) $\tan \alpha > \tan \beta$
- (b) $\tan \beta < \tan \gamma$
- (c) $\tan \gamma > \tan \alpha$
- (d) $\tan \alpha < \tan \beta$

2. From the top of a tower 100 m high a man observes the angles of depression of two ships A and B, on opposite sides of the tower as 45° and 38° respectively. If the foot of the tower and the ships are in the same horizontal line find the distance between the two ships A and B to the nearest metre. [4] (Use Mathematical Tables for this question.) Main 2023

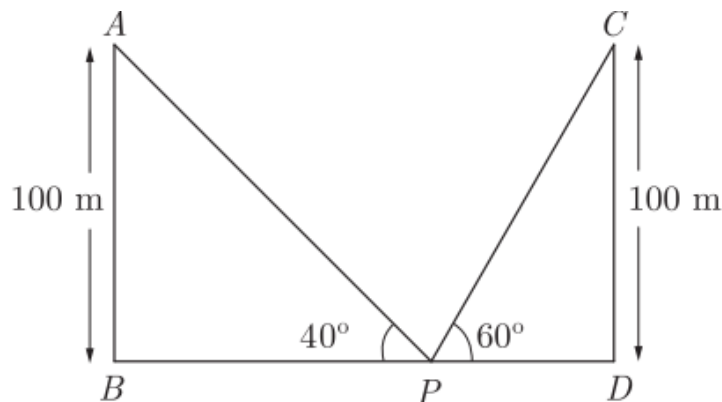


3. The angle of elevation of the top of a building from the foot of the tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60° . If the tower is 60 m high, find the height of the building.

Comp 2023

4. Two lamp posts AB and CD each of heights 100 m are on either side of the road. P is a point on the road between the two lamp posts. The angles of elevation of the top of the lamp posts from the point P are 40° and 60° . Find the distance PB and PD

Main 2022 Sem II



5. The angle of elevation of the top of a hill at the foot of a tower is 60° and the angle of elevation of the top of the tower from the foot of the hill is 30° . If the tower is 50 m high, find the height of the hill.

Comp 2022

Solution

1.

As we move away from the lamp post AB, the angle of elevation decreases.

Points are on a straight line in the order:

$$B - C - D - E$$

So the distances from the lamp post increase as:

$$BC < BD < BE$$

Hence the angles of elevation satisfy:

$$\alpha > \beta > \gamma$$

Since $\tan \theta$ increases with θ (for acute angles),

$$\tan \alpha > \tan \beta > \tan \gamma$$

✔ Correct condition:

2.

Distance of ship A from foot of tower

$$\begin{aligned}\tan 45^\circ &= \frac{DC}{AC} \\ 1 &= \frac{100}{AC} \Rightarrow AC = 100 \text{ m}\end{aligned}$$

Distance of ship B from foot of tower

$$\tan 38^\circ = \frac{DC}{BC}$$

Using tables:

$$\tan 38^\circ \approx 0.7813$$

$$\begin{aligned}0.7813 &= \frac{100}{BC} \\ BC &= \frac{100}{0.7813} \approx 128 \text{ m}\end{aligned}$$

Distance between ships A and B

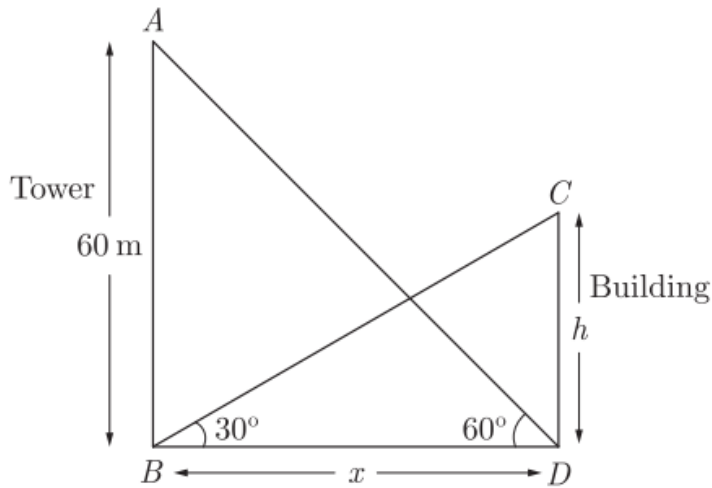
$$AB = AC + BC$$

$$AB = 100 + 128 = 228 \text{ m}$$

✓ Answer (nearest metre):

228 m

3. Let AB be the tower of 60 m height and CD be the building of h height. As per given in question we have drawn figure below.



In right $\triangle ABD$ we have

$$\begin{aligned}\tan 60^\circ &= \frac{AB}{BD} \\ \sqrt{3} &= \frac{60}{x} \\ x &= \frac{60}{\sqrt{3}} = 20\sqrt{3}\end{aligned}$$

Now, in right $\triangle BCD$ we have

$$\begin{aligned}\tan 30^\circ &= \frac{CD}{BD} = \frac{h}{x} \\ \frac{1}{\sqrt{3}} &= \frac{h}{20\sqrt{3}} \\ h &= \frac{20\sqrt{3}}{\sqrt{3}} = 20\end{aligned}$$

Hence height of the building is 20 m.

4.

Distance PB

$$\tan 40^\circ = \frac{100}{PB}$$

$$PB = \frac{100}{\tan 40^\circ}$$

Using tables: $\tan 40^\circ = 0.8391$

$$PB = \frac{100}{0.8391} \approx 119 \text{ m}$$

Distance PD

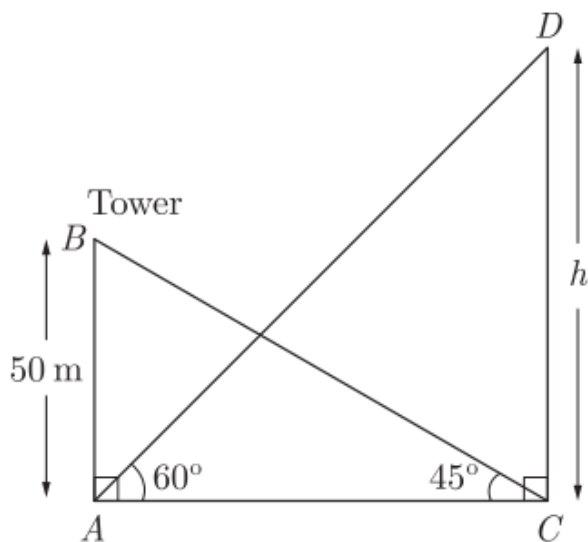
$$\tan 60^\circ = \frac{100}{PD}$$

$$PD = \frac{100}{\tan 60^\circ} = \frac{100}{\sqrt{3}} \approx 57.73 \text{ m}$$

✓ Final Answer:

Hence $PB = 119 \text{ m}$ and $PD = 57.73 \text{ m}$

5. Let AB be tower of height of 50 m and DC be hill of height h . As per given in question we have drawn figure below.



In right ΔBAC ,

$$\cos 30^\circ = \frac{AC}{50}$$

$$\sqrt{3} = \frac{AC}{50}$$

$$AC = 50\sqrt{3}$$

In right ΔACD ,

$$\tan 60^\circ = \frac{CD}{50\sqrt{3}}$$

$$\sqrt{3} = \frac{CD}{50\sqrt{3}}$$

$$CD = 50\sqrt{3} \times \sqrt{3} = 150 \text{ m}$$

Thus height of the hill $CD = 150 \text{ m}$